NUS-ISS | INTELLIGENT REASONING SYSTEMS | GROUP PROJECT

K PERFECT DAYS

Intelligent group travel planning using knowledge graphs and constraint optimization

Final Group 42 Project Report

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Project team	Student ID
Nadia Alolyani	A0354462A
Abdullah Alhussaini	A0352097Y
Sarah Almuqri	A0362910J

Executive Summary

K Perfect Days is an intelligent travel recommendation system built around a simple idea: a perfect trip should feel personal. Rather than recommending destinations based only on popularity, the system seeks to understand travelers, their interests, preferred activities, travel styles, social connections, and evolving experiences to create recommendations that become more relevant over time. The system brings together user understanding, AI-driven travel personas, Knowledge Graphs, recommendation models, and contextual reasoning to connect travelers with activities and places that fit them. It is designed not only to recommend, but to continuously learn as users interact with the product. Behind K Perfect Days is a team that transformed an ambitious concept into a working end-to-end system within a highly constrained timeframe. Each member brought a different perspective and technical strength, from data and AI to recommendation, reasoning, and system development. The result reflects extensive experimentation, problem-solving, iteration, and collaboration. More than a recommender, K Perfect Days represents the team's shared effort to explore what a truly adaptive and intelligent travel experience could become.

Our prototype combines a browser-based trip planner with a Neo4j knowledge graph, user preferences, natural-language request handling, travel-time information and a constraint-based optimizer. Rather than asking a language model to decide the entire schedule, we use it primarily to interpret the request and explain the result. The actual itinerary is selected by explicit rules and an optimization model that accounts for group preference coverage, fairness, variety, travel burden and the quality of each day.

We carried out a broad quantitative study for our solution. In the study, fairness increased the worst-served traveler's modeled preference coverage by 8.17 percentage points on average, while the persona objective improved it by 17.13 points. Travel penalties reduced modeled travel time by 20.74 minutes per itinerary. However our evaluation also found limitations including soft preferences were not being consistently fulfilled, and many venues had assumed rather than verified opening hours.

Chapter	Coverage
1. Business Case and Market Research	Problem, users, market context, alternatives, novelty and scope
2. System Design and Model	Knowledge graph, recommendation logic, rules, multi-objective optimization and architecture
3. Development and Implementation	Data preparation, application workflow, technology stack, integration, limitations and project contributions
4. Experiments and Results	Controlled quantitative evaluation and selected figures
5. Findings and Discussion	Interpretation, unexpected results, limitations and next steps
References and Appendices	Sources, module-technique mapping, installation/user guide and original proposal

1. Business Case and Market Research

1.1 The problem we set out to solve

Travel inspiration is social, but trip planning is still fragmented. A traveler might save a restaurant on Instagram, receive museum recommendations in a group chat, compare attractions on a map and ask an AI assistant for a schedule. By the time the group is ready to travel, those preferences are spread across different tools. More importantly, there is rarely a consistent way to decide whose preferences should carry more weight.

We focused on three practical problems. First, generic suggestions do not necessarily reflect what a particular traveler enjoys. Second, group planning becomes difficult when people have competing preferences or mandatory requirements. Third, an attractive list of places is not the same as a workable itinerary. Opening hours, trip duration, the distance between stops, weather and basic daily pacing all affect whether a plan can be followed.

The business question behind K Perfect Days is therefore straightforward: can we turn persistent preference information into group itineraries that are more personalized, fair and practical than a simple popularity-based list?

1.2 Why this matters now

The market evidence in our proposal suggested a clear opportunity. Phocuswright reported projected 2025 global travel gross bookings of roughly US$1.67 trillion, with online bookings exceeding US$1 trillion [1]. Expedia Group found that 61% of surveyed consumers drew travel inspiration from social media, while 73% said an influencer recommendation had influenced a booking decision [2]. Phocuswright also reported that 39% of active US travelers used AI for travel research or planning in the second half of 2025 [1].

Demand for AI assistance does not mean people want to hand over every decision. Booking.com reported that 89% of consumers wanted to use AI in future travel planning, but only 12% were comfortable with AI making decisions independently [3]. For our project, that supports a design in which recommendations and trade-offs are visible, constraints remain explicit and people can review the plan.

Travel itinerary generation is also a reasoning challenge, not only a language task. The WorldTravel benchmark describes 150 realistic scenarios with an average of more than 15 connected constraints [4]. GroupTravelBench evaluates 650 multi-person travel tasks and highlights difficulty with coordination, preference coverage and fairness [5]. We cite these studies as context, not as directly comparable scores for our own system.

1.3 Existing products and our position

Several established products already help with parts of the travel-planning journey. Mindtrip offers AI-assisted travel discovery and planning; Mapstr helps people save and discover places, including recommendations from followed maps; Polarsteps supports travel records and collaborative planning [6–8]. The gap we chose to explore is narrower and more technical: representing individual preferences as reusable graph relationships, then treating group fairness and real scheduling constraints as explicit parts of the decision.

1.4 Intended users, value, and success criteria

The main users are friends, couples, and groups organizing leisure trips to California. The planner is particularly relevant when members have different interests, when daily travel time matters or when the group has explicit requests such as visiting a museum or avoiding certain activities. For a user, the value is less manual coordination and a clearer explanation of why particular stops were

selected. For a potential travel platform, the same approach could provide a basis for personalized recommendation and group decision support.

We defined success in terms of measurable outcomes rather than a broad claim that trips are "perfect": preference coverage for every group member, least-served traveler coverage, experience variety, routing burden, fulfillment of stated requirements and schedule feasibility under the information available. Chapters 4 and 5 evaluate these measures directly.

1.5 Scope and deliberate limits

The delivered prototype and our measured experiments concentrate on California destinations, using the available place and knowledge-graph data. Venue opening times may be estimated, and the prototype should not be treated as a substitute for checking official opening hours and reservations.

2. System Design and Model

2.1 Design approach

We separated the tasks that language models handle well from those that are better expressed as rules or mathematical constraints. Natural language is useful for understanding a traveler's request and explaining a completed itinerary. However, whether two attractions can fit into the same afternoon or whether a mandatory request has been met should be checked explicitly. This separation is central to the design.

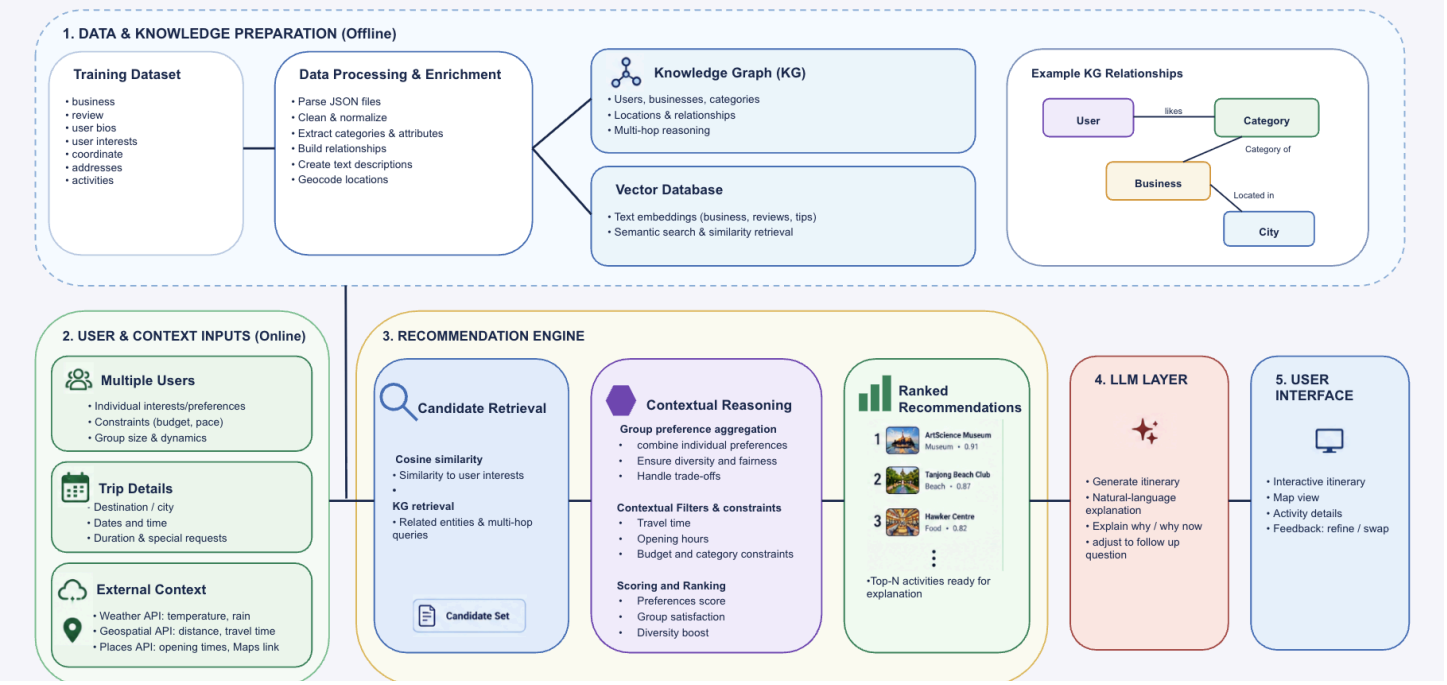


Figure 1. Implemented architecture.

2.2 Knowledge representation and personalization

Neo4j represents users, interests, activities, personas, places, place categories, and the relationships among them. User preferences are represented through interests and activities with associated confidence or preference weights, while users are mapped to the personas that best represent their broader behavioral patterns. Places are connected to categories and activities, allowing the system to translate user and persona preferences into relevant candidate places.

For a new planning request, the backend identifies the authenticated user and, for shared trips, the participating users. The planner retrieves their graph-based preference profiles and persona

relationships and uses these signals to identify and rank suitable activities and places. Candidate selection considers activity and category relevance, persona derived preferences, explicit trip requirements, candidate quality, and itinerary constraints such as location, dates, routing, and availability.

The system also represents relationships between users, such as “HAS_FRIEND” relationships, which provide the foundation for group recommendation and collaborative preference aggregation. However, because the current dataset does not contain a sufficiently complete historical user–item rating matrix, traditional collaborative filtering based on real user ratings has not been independently validated. The implemented approach is therefore more accurately described as a graph informed, persona and content-based recommendation system with group preference aggregation, while collaborative filtering remains an architectural capability for future upgrades.

2.3 Request understanding and contextual data

A natural-language request can specify destinations, number of nights, travel dates, preferences, exclusions, and mandatory activities. The request parser converts this into structured validated fields before optimization. The system can then combine the request with the persistent user preferences already held in the graph. This avoids requiring people to restate every interest whenever they plan another trip.

The itinerary pipeline uses openrouteservice routing matrices to estimate travel times, Open-Meteo for weather context, and Geoapify place information for opening hours. Weather forecasts have a limited forward window, so longer-range plans may use historical context.

2.4 Rules and optimization

The planner first builds candidate places and possible daily tours. It checks feasibility at the tour-building stage. Then our optimization engine, we choose Google OR-Tools CP-SAT, selects a compatible option for each day. This is much more controlled than asking a language model to write an unrestricted schedule and hoping the timings work.

Among the feasible options, the objective rewards overall group preference match, the minimum preference coverage of the least-served traveler, stated optional wishes, activity variety and daily quality. It penalizes longer modeled travel. Increasing variety can reduce preference coverage, while aggressively maximizing one person’s fit can leave others with fewer appealing stops. As such, all the elements need to work together and be optimized simultaneously. The factorial experiments in Chapter 4 are designed to reveal those trade-offs.

Hard requirements include feasible visiting windows, suitable daily sequencing, avoiding excluded categories or places, avoiding repeated locations, and satisfying explicitly mandatory activities when candidates exist. Where a mandatory request cannot be satisfied, the system should reject the plan rather than invent a stop. The model also applies day-composition rules to discourage unrealistic or nearly empty sightseeing days.

2.5 Integration of IRS Techniques

IRS technique area	How we applied it
Knowledge-based reasoning	Neo4j entities/relationships, preference reasoning, typed trip requests and rule checks
Business resource optimization	CP-SAT selection of feasible daily tour combinations under hard and soft requirements
Knowledge discovery and recommendation	Graph-informed candidate retrieval, persona scoring and feedback signals
Cognitive techniques	Natural-language parsing and readable explanation of structured itinerary decisions

3. System Development and Implementation

3.1 Data preparation and the knowledge graph

The proposal initially considered public OkCupid data to generate synthetic user interests, alongside Yelp and Kaggle as open-source sources for activities and attractions. During implementation, Yelp and Kaggle were found to provide insufficient coverage and depth for the required activities and places. Therefore, these sources were replaced with data retrieved through Google's API to provide a richer and more relevant activity and place dataset for California. OkCupid data remained the foundation for inferring travel profiles, which were generalized into personas to address data noise and the user cold-start problem.

Within the Knowledge Graph, users were connected to personas, interests, activities, places, and other users through defined relationships. Personas provided a generalized preference layer, while explicit user interactions were designed to continuously refine persona alignment at runtime. Friendship relationships were also incorporated to support collaborative filtering for group recommendations.

Due to the absence of a sufficiently complete real user rating dataset, credible rating prediction MAE or leave-one out collaborative filtering accuracy could not be calculated. Evaluation therefore focused on measurable system behavior and acknowledged this limitation explicitly.

3.2 Frontend and backend integration

The user interface is built with HTML, CSS and JavaScript; it is styled with nature-inspired colors that we found appropriate for this kind of application. The backend in `scripts/server.py` handles session-based authorization, account operations, saved preferences and shared trips. For an itinerary request, it validates the logged-in account, resolves the relevant member profile IDs and submits an authenticated planning job.

The separate `scripts/planner_jobs.py` process executes `scripts/planner_engine.py`, stores each run's log and private outputs, and makes the result available through job polling. This keeps a long optimization request from blocking a browser response. Completed results are returned as a structured daily plan, metrics and a written itinerary. Jobs are kept in process memory, so an application restart loses the in-memory job registry even though output files remain on disk.

Implemented component	Where it lives	Role
Web pages and interaction	scripts/index.html, app.js, styles.css	Signup, preferences, Discover, shared voting, trip planning and itinerary display
User graph and feedback	scripts/user_service.py	Neo4j user records, activity interests, feedback and persona links
HTTP APIs and authorization	scripts/server.py	Authenticated requests, sessions, shared trips and planner endpoints
Background execution	scripts/planner_jobs.py	Run optimizer as a separate job and return saved results
Reasoning and optimization	scripts/planner_engine.py	Request parsing, graph/candidate data, routing, hard rules and CP-SAT optimization

3.3 End-to-end user journey

A typical user starts by creating an account and choosing preferred activities so they are mapped to one of our pre-defined persona. Afterwards, they can plan either a solo or a group trip by picking a destination within California and the dates of their trip. The system will then automatically generate a prompt based on the user's data and feed the prompt to our LLM which prints out an AI-generated travel itinerary. Travelers can interact with the itinerary and LLM to customize the trip to their linking; they can also replace some of the suggested places with a similar alternative.

For group trips, the original user generates an invite link and shares it with the trip members. Members joining through the link can create an account to pick their interests then they are automatically taken to the group trip itinerary page. Every time a new user joins the trip, the itinerary is updated to include his interests and its updated for all people in the same trip. For group trips, attendees have the option to like or replace recommended places, if at least half the attendees liked a place, the place is kept; by the same logic, if at least half the attendees asked for a place to be replaced, then said place is removed. Finally itinerary is then saved under the Saved Trips page.

All users can view previous trips under the Saved Trips page.

3.4 Implementation challenges and decisions

The biggest engineering challenge was integration. The recommendation interface and user model were developed separately from the detailed itinerary notebook, and the two systems did not originally share the same runtime or input format. Extracting the planner into an executable Python engine, passing authenticated profile IDs into it and creating a job-based backend were therefore important development steps.

We also faced practical issues with external data and solver complexity. Routing and opening-hours services can return incomplete information, and availability assumptions affect what the solver considers feasible. Large trips generate many candidate tours, making proof of mathematical optimality harder even when a valid itinerary is found.

3.5 Demonstration and evidence of contribution

The course assessment also requires two short demonstration videos: one explaining the user problem, value and product use case, and another presenting the system design and technical reasoning. The report documents the implemented workflows and tested methods that can be demonstrated in those recordings. Video production and individual peer reviews are separate submission items; this report does not claim they have already been recorded or submitted. Consistent with the course instructions, any presentation speech should be delivered by the students rather than generated as AI speech.

4. Experiments and Results

We evaluated K Perfect Days to find out whether the main reasoning methods actually improved the travel plans. Producing a valid itinerary was only the starting point. We wanted to see whether personalisation, fair treatment of group members, activity variety and shorter travel times made a measurable difference, and where the system struggled.

A short guide to the reported measures

Measure	How to read it
Lowest traveler preference match	The lowest score across the group according to our own model; higher means the least-served member is better covered.
NDCG@K	A ranking score that gives more credit when highly relevant suggestions appear near the top K.
OPTIMAL versus FEASIBLE	Both follow the encoded rules. OPTIMAL means the solver proved it cannot find a better answer; FEASIBLE has no such proof.
Assumed hours	An opening window supplied by the planner when reliable date-specific venue evidence was not available.

4.1 How we tested the system

We tested our project using traveler profiles from Neo4j, candidate activities and route estimates, and the constraint solver that builds the final daily schedule. Our main experiment compared the full planner with six versions in which one scoring component was switched off. Each version used the

same candidate places, travel estimates and scheduling rules within a trip. This means the differences mainly reflect the scoring choices rather than a different set of inputs.

Study	Coverage	What we measured
Main optimisation study	23 trips × 7 variants = 161 runs	Preference fit, fairness, variety, travel, schedule validity
Sensitivity	42 runs	Changes in candidate-pool size and travel date
Recommendation and language	312 LLM relevance labels, 119 itinerary ratings, 4 prompts	Ranking quality, simulated satisfaction, request parsing
Solver and request tests	9 solver-budget trials, 30 request cases	Long-trip performance, mandatory and optional wishes
Opening-hours audit	414 places	How much of the schedule relied on assumed hours

The 23 main trips covered 2, 4 and 7 nights, groups of 1, 2 and 4 travelers, and one or two cities, along with several additional examples. Most trips used San Francisco, Los Angeles and San Diego. We used a 30-second solve limit for the main study.

For group outcomes, we report the average predicted preference match and the score of the least-served traveler. These are based on our own recommendation model. We also measure the number of different experience types and estimated time spent traveling. For recommendation ranking, we use Precision@5 and NDCG@5/10, which indicate how often relevant suggestions appear near the top.

4.2 What happens when we remove parts of the optimiser?

This was our most useful controlled experiment. The table below compares the full model with versions that remove a single scoring component. Positive or negative outcomes are reported even when they do not favor our final design.

Component tested	Measured effect of full model	Takeaway
Fairness	+8.17 percentage points in the lowest traveler preference score	Helps avoid leaving one traveler behind
Persona preferences	+17.13 points in the lowest traveler preference score	Largest gain in modeled preference matching
Variety	+0.30 different experience types per trip	More variety, with a small preference and travel cost
Travel penalty	20.74 fewer estimated travel minutes per trip	Less unnecessary movement
Day composition	No meal-only ordinary days versus 16 without the rule	Better daily balance, but less modeled preference fit
Soft-wish reward	No aggregate change in the initial study	Needed a more targeted test

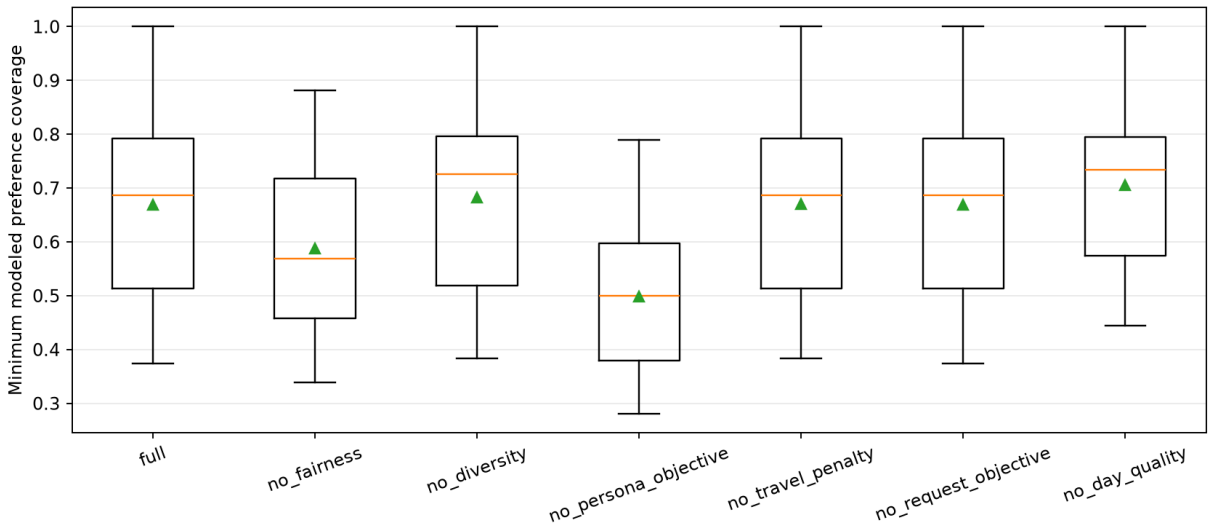


Figure 2. Lowest traveler preference match across the seven objective versions. The persona-free objective performed worst on average.

Across the matched trips, the fairness improvement had a descriptive 95% bootstrap interval of 4.53 to 12.30 percentage points. The corresponding range for the persona effect was 12.71 to 22.04 points. Fairness improved the lowest-scoring traveler in 15 of 23 trips and tied in eight. Looking only at trips with at least two people, the average benefit was still 5.48 percentage points. The persona term had an even larger effect, with an increase of 17.13 points in the lowest traveler preference score

The differences are meaningful for the model we built, but not every goal improves together. Adding variety gave travelers more different kinds of activities while reducing minimum modeled preference fit by around 1.25 points. Removing day-quality rules improved numerical preference scores but produced poorly balanced days, including some with meals and little else. Those trade-offs support keeping multiple objectives rather than optimizing only one number.

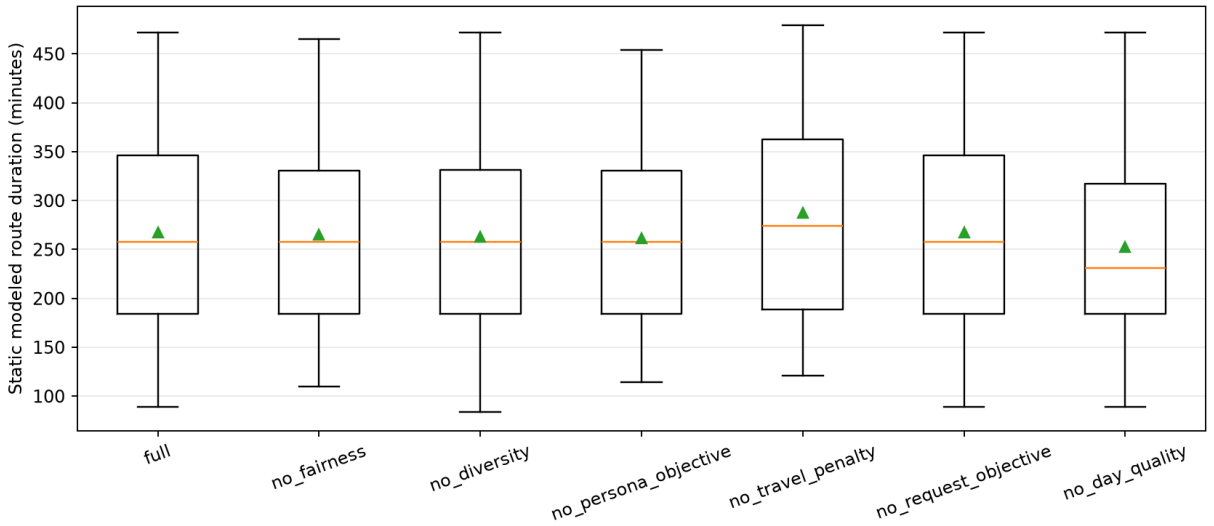


Figure 3. Estimated travel time across objective versions. Turning off the travel penalty increased travel on average.

For the travel results, the same types of trips were planned with about 21 fewer minutes of movement when travel time was rewarded in the scoring system. This is based on static route estimates, not live traffic. The comparison nevertheless shows that the routing objective has a practical effect.

4.3 Can the planner handle harder trips?

All 161 main runs returned valid schedules under the rules programmed into the solver. Of those, 140 were proven optimal and 21 were feasible solutions that the solver could not prove were the best possible within the time limit. All 21 came from three seven-night, single-city cases. Interestingly, seven-night trips across two cities were solved optimally. This shows that difficulty depends on the combinations of possible daily activities, not only the number of cities or travelers. This is mainly due to combinatorial explosion when we have a large number of activities and permutations in a city

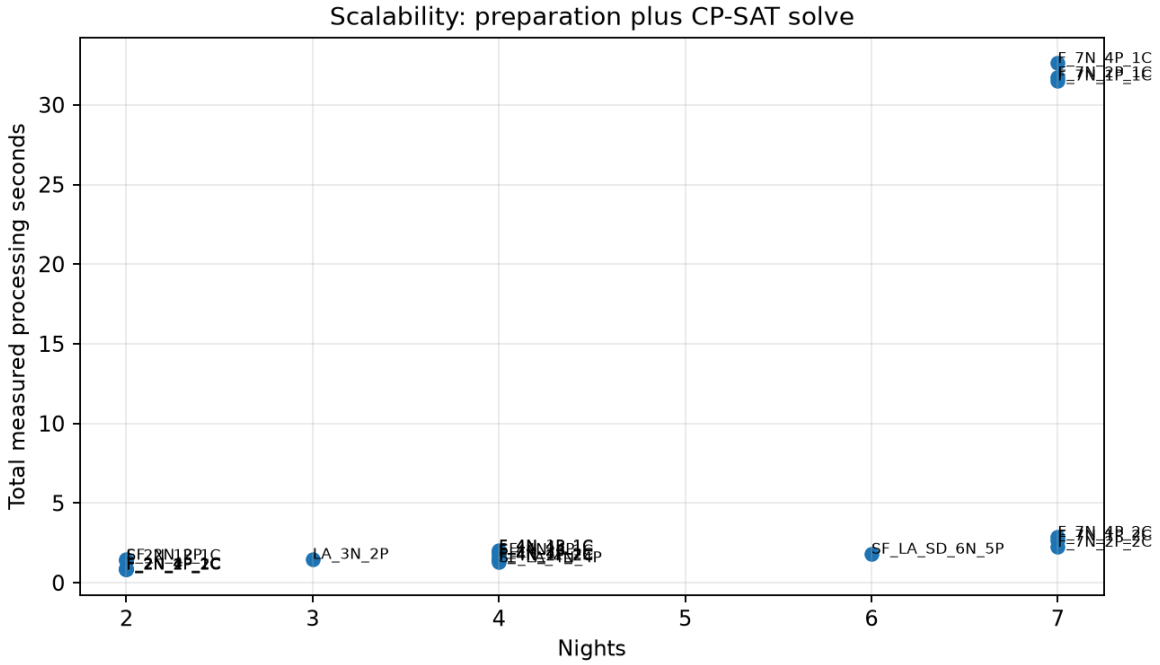


Figure 4. Runtime becomes much harder to predict on longer trips, especially seven-night single-city schedules.

We also changed the number of candidate places (16, 24 and 34) and compared two trip dates. All 42 sensitivity runs were optimal. For the November San Francisco trip, the itinerary stayed the same even as more places became available. For the April trip, more choices changed the schedule and did not consistently improve preference matching. In other words, offering the solver more candidates does not guarantee a better result on every measure.

4.4 Do the recommendations seem relevant?

Without a suitable dataset of actual user ratings, we asked an LLM to grade 312 candidate places on a 0 to 3 relevance scale, using stated traveler interests. We compared ranking by persona fit with a simple ranking that favored well-known places.

Ranking metric	Persona ranking	Famous-place baseline
Precision@5 (relevant in top five)	0.477	0.385
NDCG@5 (relevance near top)	0.529	0.265
NDCG@10 (relevance near top ten)	0.643	0.367

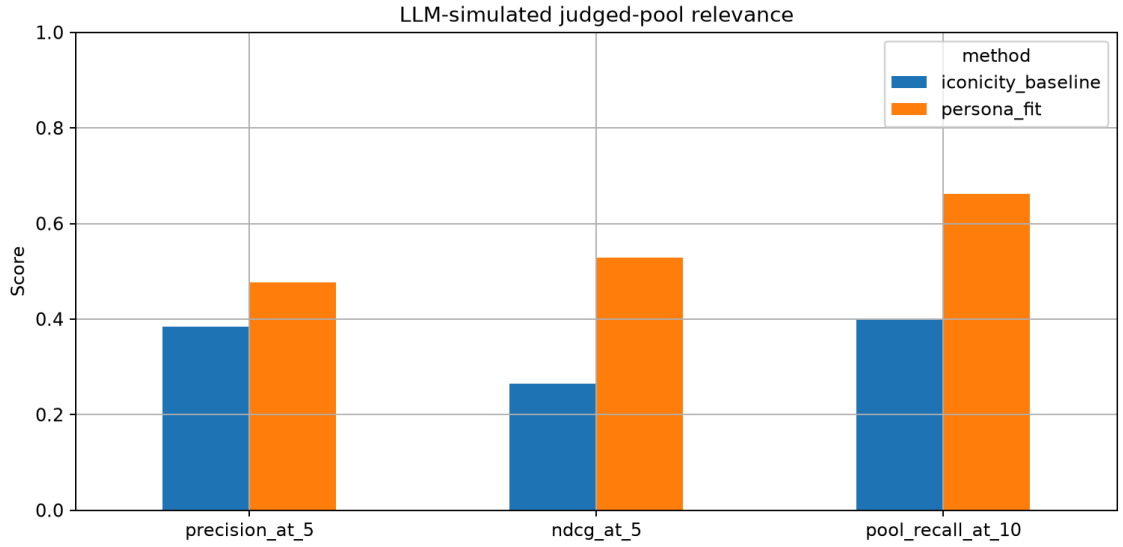


Figure 5. LLM-judged recommendation results. The persona method ranked relevant activities better on average.

The persona method did better on average, especially when ranking the top ten. However, it beat the baseline in only seven of the 13 pools on NDCG@5, and it was not consistently better at finding the very first relevant place. The result is promising, but not universal.

4.5 Would travelers enjoy the complete itinerary?

We used an LLM to act as a reviewer for 119 traveler-itinerary rating records across five types of trips. This was a practical way to get consistent feedback during development.

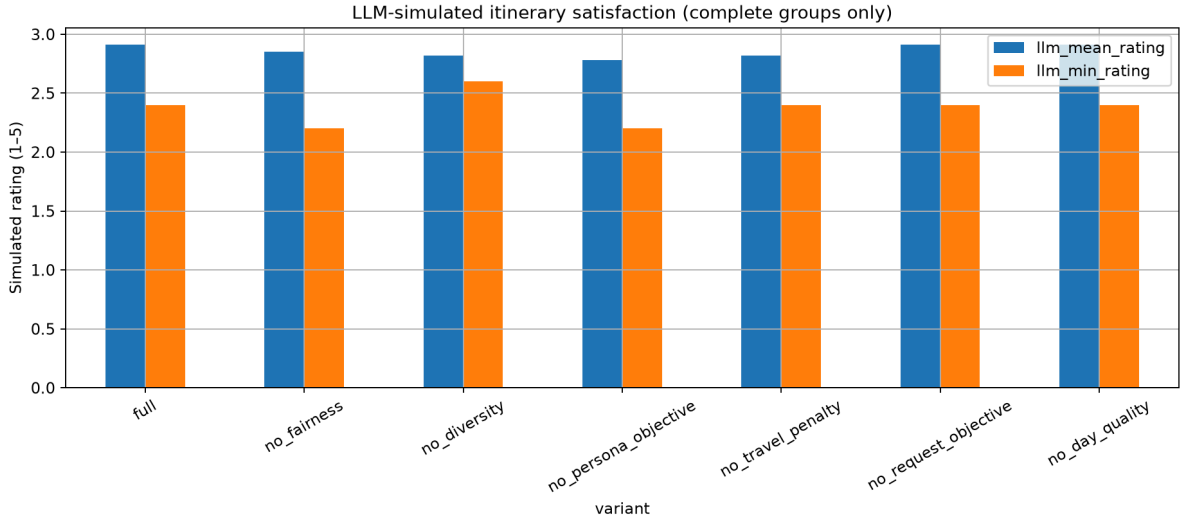


Figure 6. Simulated satisfaction with complete itineraries. The full model performs slightly better on the average rating, but not on every comparison.

The full model averaged **2.91 out of 5**, compared with 2.85 without fairness and 2.78 without persona scoring. Those differences are small. Removing diversity actually improved the average rating of the least-satisfied group member in this small sample (2.60 instead of 2.40), however as stated, it leads to worse quality days.

4.6 Solver time

We retested the three difficult seven-night, single-city problems with time limits of 30, 60 and 120 seconds, using the original seed and the same feasible tour options. All nine runs found valid solutions, but none proved optimality. The chosen solution quality did not improve when we gave the solver more time. Repeating a 30-second run did not always select exactly the same itinerary, even

with the same seed. We therefore report these longer cases as valid plans, but not mathematically proven best plans. As such making the time limit much longer won't support much in terms of results but might turn away potential users.

4.7 Would the places actually be open?

This exposed the largest practical weakness. We checked 414 scheduled visits from 37 generated itineraries, covering 56 distinct place IDs. 321 visits (77.5%) used assumed opening hours and access. The remaining 93 relied on typical weekly hours stored in the planner, which also were not confirmed for the exact future dates.

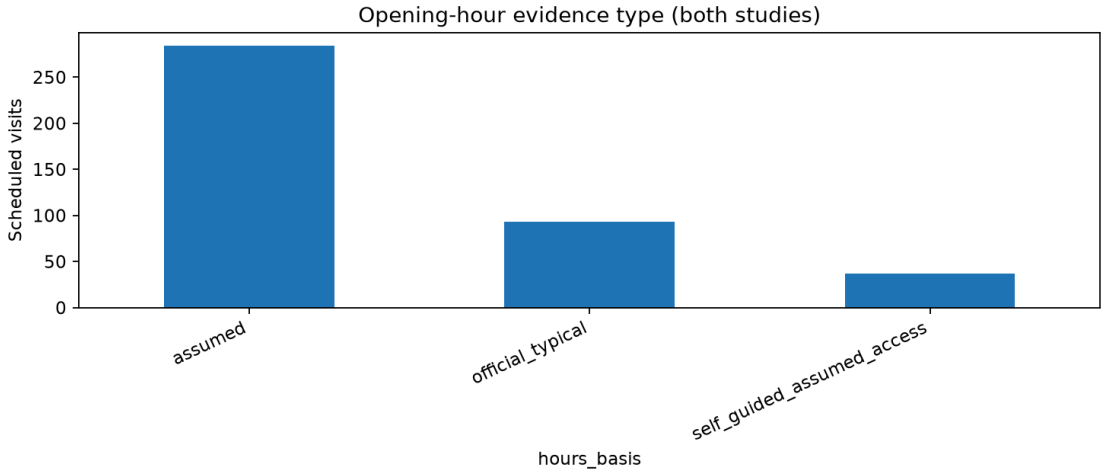


Figure 7. Where the scheduled hours came from. Most visits depended on assumed hours or access.

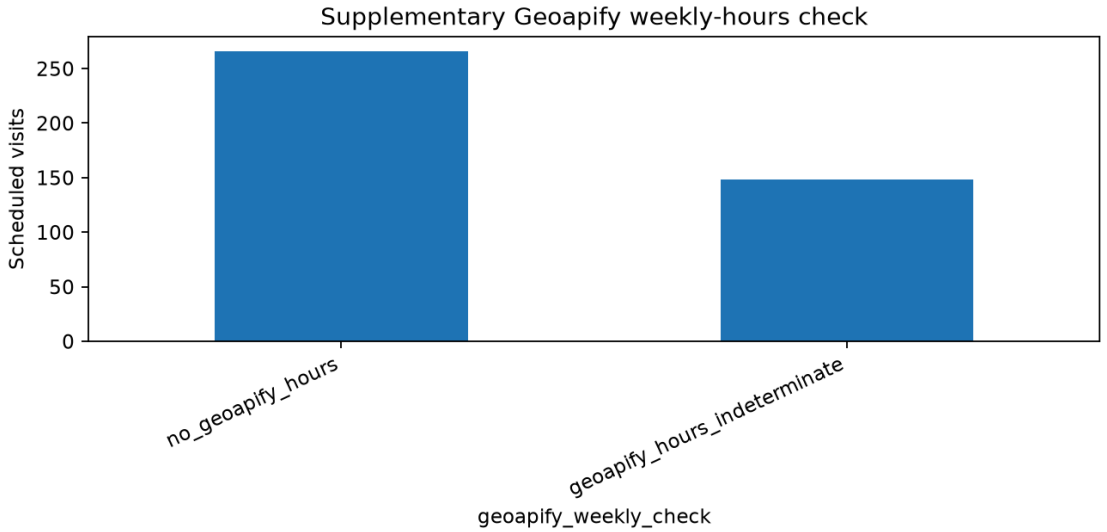


Figure 8. Weekly opening-hour data were unavailable for most audited visits. This is missing evidence, not proof of closure.

Geoapify supplied usable weekly opening hours for 148 visits. A supplementary check flagged seven possible mismatches, involving Rich Table, Nightbird and Griffith Observatory. Our later review of venue information suggested that three Nightbird visits had credible weekly-hours concerns, three Rich Table flags conflicted with current official information, and one Griffith Observatory case depended on whether indoor access was needed. None of these checks confirmed date-specific closures or bookings.

5. Findings and Discussion

Taken together, the experiments give us a useful picture of where K Perfect Days works and where it still needs improvement. The central approach is sound: we can combine traveler profiles, route information and hard scheduling rules into a working group itinerary. The bigger question is how much that numerical success translates into a trip people would actually want to take.

5.1 What worked, what was mixed, and what did not

Assessment	What we learned
Strong	Fairness protected the lowest-scoring traveler; persona scoring improved preference fit; the travel term reduced estimated movement. These effects were measurable in matched comparisons.
Mixed	Variety and good day composition improved the shape of the itinerary but sometimes reduced modeled preference fit. Optional wish matching improved only slightly.
Weak	Long single-city trips were difficult to prove optimal. Most hours could not be verified for the scheduled dates.

We did not just switch features on and off and compare unrelated trips. Every optimisation version saw the same candidate pool and constraints, which made the fairness, persona and routing differences easier to interpret.

The findings also highlighted valuable opportunities for improvement. The moderate LLM ratings suggest that activity selection and pacing still need refinement. The soft-request tests show that an optional preference may be missed even when the request has been captured correctly. The hours audit showed that a technically valid schedule can still contain uncertain real-world details. These are practical problems a traveler would notice immediately.

5.2 What this means for the project

From a technical standpoint, we would keep the current knowledge-graph and constraint-optimisation approach. The controlled tests show that these pieces contribute something useful. However, spending more CPU time alone did not solve our difficult seven-night examples. Better candidate selection, more complete traveler preferences and improved real-world constraints are likely to matter more.

From a product standpoint, K Perfect Days is a promising prototype, not yet a travel assistant we would trust without checking the details.

5.3 Limitations and next steps

We would focus future development on four areas. First, collect a small set of genuine traveler ratings and compare recommendations against held-out preferences. Second, verify opening hours, bookings and intercity transport for the actual travel date before confirming an itinerary. Third, improve how the planner handles harder optional wishes and balance between experience variety and personal fit. Fourth, test complete itineraries with people, including pacing and journey time, rather than relying only on proxy scores.

There are also limits to the evidence itself. The scenarios were deliberately constructed rather than randomly sampled, several shared the same people and cities, and many route requests used cached values. The reported bootstrap intervals describe variation across these test scenarios, not across the wider traveling public. Likewise, a solver result marked FEASIBLE confirms that the generated itinerary satisfies the defined scheduling constraints, providing a valid and practical solution. Rather, than providing the best possible itinerary.

5.4 Overall conclusion

The experiments support the main technical idea behind K Perfect Days. Personalisation, fairness and routing each made a measurable difference, and the planner consistently built schedules that met its

programmed hard rules. But they also showed that numerical optimisation alone is not the same as planning a great trip. Our next improvement should be to strengthen the underlying travel information and validate the experience with real users, while building on the strengths of the existing reasoning framework."

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Appendix A. Mapping the System to IRS Techniques

The Practice Module asks teams to demonstrate at least three of four technique groups. We mapped the delivered components against all four, while distinguishing the implementation from future ideas.

Module / technique	Evidence in the final project	Implementation qualification
Machine Reasoning: knowledge graphs	Neo4j user, persona, activity, place and category relationships; graph lookups	Used to retrieve structured user and place evidence
Reasoning Systems: rules and constraints	Hard request checks, place exclusions, activity windows, schedule consistency	Feasibility depends on the accuracy of hours and external information
Reasoning Systems: optimization	OR-Tools CP-SAT selects multi-day tour combinations with group/fairness/travel objectives	Some larger runs yielded valid FEASIBLE plans without proven optimality
Recommendation and knowledge discovery	Persona/content scoring, user interest updates and Top-K evaluation	No independently validated user-rating collaborative-filtering dataset
Cognitive Systems: language	OpenAI request extraction and final explanation from selected schedules	LLM proxy judging is not a genuine human-user study
System integration and usability	Browser interface, authenticated API, shared-trip voting and asynchronous planner jobs	Full conversational re-planning is not yet implemented

Our contribution is the integration of these methods into a runnable system and the evaluation of their separate effects through controlled tests. The experiment files and original source code provide the reproducibility trail.

Appendix B. Installation and User Guide

B.1 Requirements and configuration

Use Python 3.11 or 3.12 and a Neo4j database populated with the appropriate user, persona and place graph. The planner also needs an OpenAI API key and an openrouteservice key. Geoapify is optional; Open-Meteo does not require a personal key for the tested usage. API accounts may incur charges.

From the extracted repository root, create a virtual environment and install the listed dependencies:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r scripts/requirements.txt
```

Create scripts/.env using your own credentials and the following variable names. Never commit real keys or passwords to GitHub:

```
NEO4J_URI=neo4j+s://YOUR_AURA_HOST
NEO4J_USERNAME=neo4j
NEO4J_PASSWORD=YOUR_PASSWORD
NEO4J_DATABASE=neo4j
OPENAI_API_KEY=YOUR_OPENAI_KEY
ORS_API_KEY=YOUR_ORS_KEY
GEOAPIFY_API_KEY=OPTIONAL_KEY
```

B.2 Start and use the system

1. Run `python scripts/server.py` from the repository root.

2. Open http://127.0.0.1:8000/scripts/ in a browser.

3. Sign up or log in, choose activities, and explore the Discover recommendations.

4. Optionally create a shared trip, invite other users, and collect votes.

5. In Plan a trip, type a complete request with city, travel dates or number of nights, preferences and hard exclusions.

6. Select Generate itinerary, review the day-by-day result, and check warnings about unverified hours.

B.3 Troubleshooting and known limitations

Issue	What to check
Server does not start	Verify Python packages, environment file and Neo4j credentials; check both Neo4j URI variables
Planning job fails	Inspect scripts/planner_results/<job_id>/run.log for API, database, routing or solver errors
Trip returns few places	Confirm graph place coverage, destination support and mandatory requests
A venue may be closed	Verify the venue's official current hours and booking details before traveling
Chat does not change recommendations	The final repository contains a placeholder recommendation chat panel
Job result lost after server restart	Current job registry is in memory; previous output files are still stored privately

Security note: the supplied repository snapshot contains a scripts/.env file and local account data. Do not publish those files. Rotate any exposed keys before submitting a public repository, and keep account and output files out of version control.

Appendix C. Original Project Proposal

The original ten-slide K Perfect Days proposal is reproduced on the following pages. Its Tokyo walkthrough and sample scores were clearly labeled illustrative.

Original proposal | Slide 1 of 10



Original proposal | Slide 2 of 10

IRS PRACTICE MODULE | PROJECT PROPOSAL

K PERFECT DAYS

K Perfect Days

A social travel planning platform that learns each traveler's tastes then reasons across the people going to build their K perfect days itinerary.

INTRODUCTION

THE PROBLEM

Travel inspiration is social. Trip planning is fragmented.

Today, travelers discover places through friends, creators and social feeds but planning still happens across saved posts, maps, group chats and generic AI tools.

01 **Preference history gets lost**
One-off prompts do not capture how a traveler actually behaves over time.

02 **Groups create trade-offs**
Different people have conflicting tastes, must-haves and constraints.

03 **Good recommendations can be bad plans**
Opening hours, distance, budgets and weather can make a "great" POI infeasible.

THE SOLUTION

Turn social travel behavior into persistent preference intelligence then optimize the trip for the group.

HOW IT WORKS

1 **Travel Graph**
- Visited / saved / rated places
- Interests & categories
- Budget, pace, timing
- Dietary / accessibility

2 **Social Graph**
- Friends & followers
- Creators / influencers
- Taste similarity & trust
- Social recommendation signals

3 **Group Context**
- Choose trip companions
- Parse trip prompt
- Aggregate each profile
- Identify conflicts & must-haves

4 **Intelligent Reasoning System**
- Collaborative + content-based filtering
- KG reasoning
- Rules & constraints engine
- Multi-objective optimization

5 **Learn & Improve**
- Explain every choice
- Accept / swap / vote
- Rate places post-trip
- Update graphs for next trip

PROJECT GOALS

01 **Learn**
Build evolving user/social preference graphs from ratings, visits and saves.

02 **Recommend**
Improve relevance, personalization and diversity using hybrid/social recommendations.

03 **Reason**
Balance multiple travelers fairly while enforcing explicit rules and live context.

04 **Optimize**
Generate feasible K-day schedules and quantify improvements against simpler baselines.

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K PERFECT DAYS

Walkthrough: from @s to an explainable group itinerary

Illustrative end-to-end example: preferences are retrieved from persistent knowledge graphs, while the prompt only supplies trip-specific constraints.

EXAMPLE

USER PROMPT

"Plan 3 perfect days in Tokyo for me, @Nadia and @Sarah. We arrive Friday 14:00 and leave Monday 18:00. Halal meals only. No alarms before 09:30. Max ¥30,000 per person/day. One splurge dinner. Keep Saturday night social. Cap walking at ~20k steps/day."

1 WHAT THE SYSTEM ALREADY KNOWS

Illustrative learned signals — not stated in the prompt

@Abdullah — 47 ratings • 22 saves

food markets • gaming / tech • skyline views

High similarity with @TokyoEats

@Nadia — 36 ratings • 18 saves

art / design • local neighborhoods • desserts

Often agrees with @CityCurator

@Sarah — 41 ratings • 27 saves

fashion • nature • immersive experiences

Strong overlap with @HiddenTokyo

Social signal ≠ popularity. The model can weight a friend / creator more when historical ratings show higher taste similarity.

2 REASON IN REAL TIME

LIVE CONTEXT FETCH

Illustrative forecast: Sat 15:00–21:00 rain risk 70% → push outdoor-heavy activities to Sunday.

A — Candidate generation

Collaborative + content + social-trust signals produce a personalized POI pool.

B — Rule filtering

Remove anything violating dietary restrictions, time preferences, opening hours, budget, distance or any hard-set user rules.

C — Group-fair optimizer

Maximize group satisfaction + fairness + diversity; minimize travel time + cost; target zero hard violations.

D — Dynamic re-optimization

Weather / delays / attendee changes trigger a new feasible plan without losing the group objective.

OPTIMIZE: satisfaction + fairness + diversity − travel time - cost

3 ILLUSTRATIVE FINAL RESULT

FRI — Arrival + city energy

16:30 Shibuya orientation
19:30 Halal dinner cluster
21:15 Skyline / social stop

SAT — Rain-adapted indoor day

10:45 teamLab Borderless
13:30 Halal lunch
15:30 Akihabara gaming / arcades
20:30 Shinjuku social night

SUN — Dry-weather neighborhoods

10:30 Asakusa + old Tokyo
13:15 Halal lunch
15:00 Harajuku / Meiji area
18:30 Sunset viewpoint

WHY THIS PLAN? Rain-aware sequencing • every traveler gets a high-affinity anchor • geography clustered • hard rules satisfied.

Example only — itinerary and scores are illustrative, not experimental results.

Group fit	Fairness	Diversity	Hard violations
91/100	0.86	0.79	0

Example metrics

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IRS PRACTICE MODULE | PROJECT PROPOSAL

K PERFECT DAYS

Project Background / Market Context

Travel discovery has gone social and planning has gone AI, but turning multiple human preferences into a fair, feasible, and live itinerary remains a real reasoning problem.

WHY NOW

$1.6T
Global travel market context in 2026; online travel bookings have already surpassed $1T. [1]

>60%
Consumers citing social media as a source of travel inspiration in 2025; 73% say influencer recommendations affected booking decisions. [2]

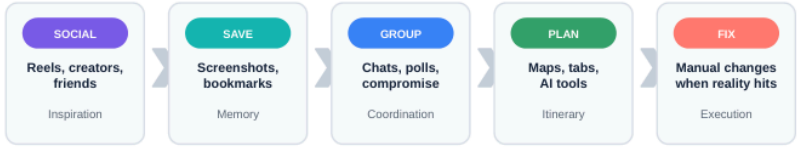
39%
Active U.S. travelers using AI for travel research/planning in 2H25; 58% among Millennials. [1]

89%
Consumers wanting to use AI in future travel planning, but only 12% are comfortable with AI deciding independently. [3]

15+
Average interdependent constraints per WorldTravel scenario, showing why realistic itinerary feasibility is non-trivial. [4]

WHAT BREAKS TODAY

The journey from inspiration → group agreement → executable itinerary is fragmented and largely stateless.



- **Personalization is shallow** — One-off prompts and popularity lists do not exploit long-term ratings, visits or social-taste relationships.
- **Group preference conflicts are under-modeled** — Collaboration tools collect opinions, they do not necessarily optimize fairness across travelers.
- **Recommendations ≠ feasible schedules** — A liked place can still fail because of time, distance, budget, opening hours or weather.

WHY THIS NEEDS INTELLIGENT REASONING

WORLDTRAVEL [4]

150 real-world scenarios

Where GPT-5.2 reached only 32.67% feasibility in text-only settings.

GROUPTRAVELBENCH [5]

650 multi-user tasks

Frontier models still show substantial weakness in preference coverage and group fairness.

THE OPPORTUNITY

K Perfect Days targets the gap between "AI that can suggest places" and an explainable decision system that can learn people, arbitrate a group, enforce rules, react to weather and optimize an executable schedule.

Sources: [1] Phocuswright, Travel Forward: Data, Insights and Trends for 2026. [2] Expedia Group, 2025 Traveler Value Index. [3] Booking.com, Global AI Sentiment Report (2025). [4] Wang et al., WorldTravel, arXiv:2602.08367 (2026). [5] Cheng et al., GroupTravelBench, arXiv:2605.25200 (2026).

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K PERFECT DAYS

Market Research

The category is validated, but “AI + social travel” is not enough. Our space is explicit social taste modeling + multi-user fairness + constraint optimization + explainability.

MARKET LANDSCAPE

PUBLICLY POSITIONED CAPABILITIES — SEP 2026

✓ verified on public product material △ partial / adjacent — not publicly specified

Capability	Mindtrip	Roameo	Mapstr	Polarsteps	K Perfect Days
AI itinerary planning	✓	✓	—	✓	✓
Friends / group collaboration	✓	✓	△	△	✓
Creator / social discovery	✓	△	✓	✓	✓
Persistent preference history	△	△	✓	✓	✓
Taste-weighted social recommendation	—	△	✓	—	✓
Explicit group fairness objective	—	—	—	—	✓
Hard constraints + weather re-optimization	△	△	—	△	✓
Explainable provenance / counterfactuals	△	—	△	—	✓

Competitor features are based on public product pages; “—” means not publicly specified, not proof of absence.

HOW WE DIFFER FROM MINDTRIP

Mindtrip is the closest mainstream comparator: it already offers conversational personalization, creator guides, collaborative planning and recommendations that “balance everyone’s vibe.” [6]

1. **Explicit Social Travel KG**
   KPD makes people, places, ratings, saves, follows and taste-similarity links queryable reasoning objects, not just profile text.
2. **Who should I trust?**
   Recommendations can weight friends / creators by measured taste affinity, not follower count or generic popularity.
3. **Fairness is an objective**
   KPD optimizes overall utility & minimum individual satisfaction so one traveler is not repeatedly sacrificed.
4. **Rules + weather drive the plan**
   Hard constraints are enforced before optimization. Weather can trigger automatic re-sequencing and re-optimization.
5. **Explainable social provenance**
   Show why a POI won, whose historical signal mattered, and how the plan changes if a traveler / budget / weather assumption changes.

POSITIONING: not another AI planner but a social graph-powered GROUP DECISION SYSTEM for travel.

Sources: [6] Mindtrip Traveler FAQs + Creator FAQs (accessed Sep 2026). [7] Roameo AI Google Play listing (updated Aug 25, 2026). [8] Mapstr Matching + Mapstr official features. [9] Polarsteps official product pages. Supporting research: GroupTravelBench (2026), WorldTravel (2026).

Data Collection

The model will be trained on synthetic user data from okcupid and validated with both synthetic and authentic data collected from real-life users. Recommendations for places to visit will be pulled from yelp and kaggle for United States locations only

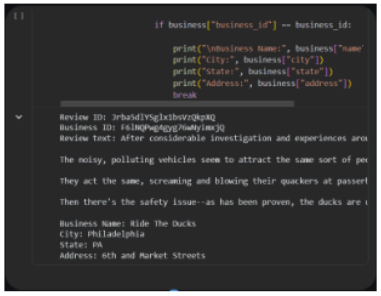
okcupid

okcupid public user data will be used to generate synthetic users and build knowledge graphs based on okcupid user interests



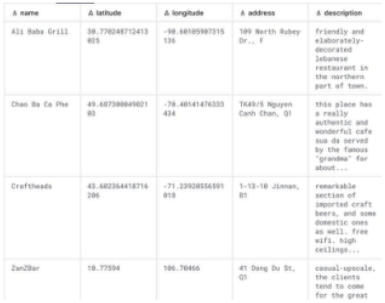
yelp

yelp data will be used to recommend locations and businesses based on the user's dietary restriction and interests.



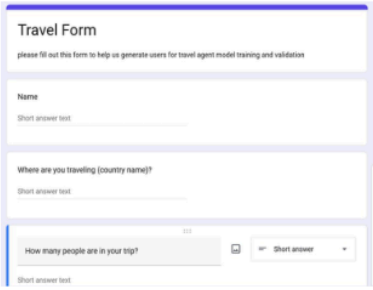
kaggle

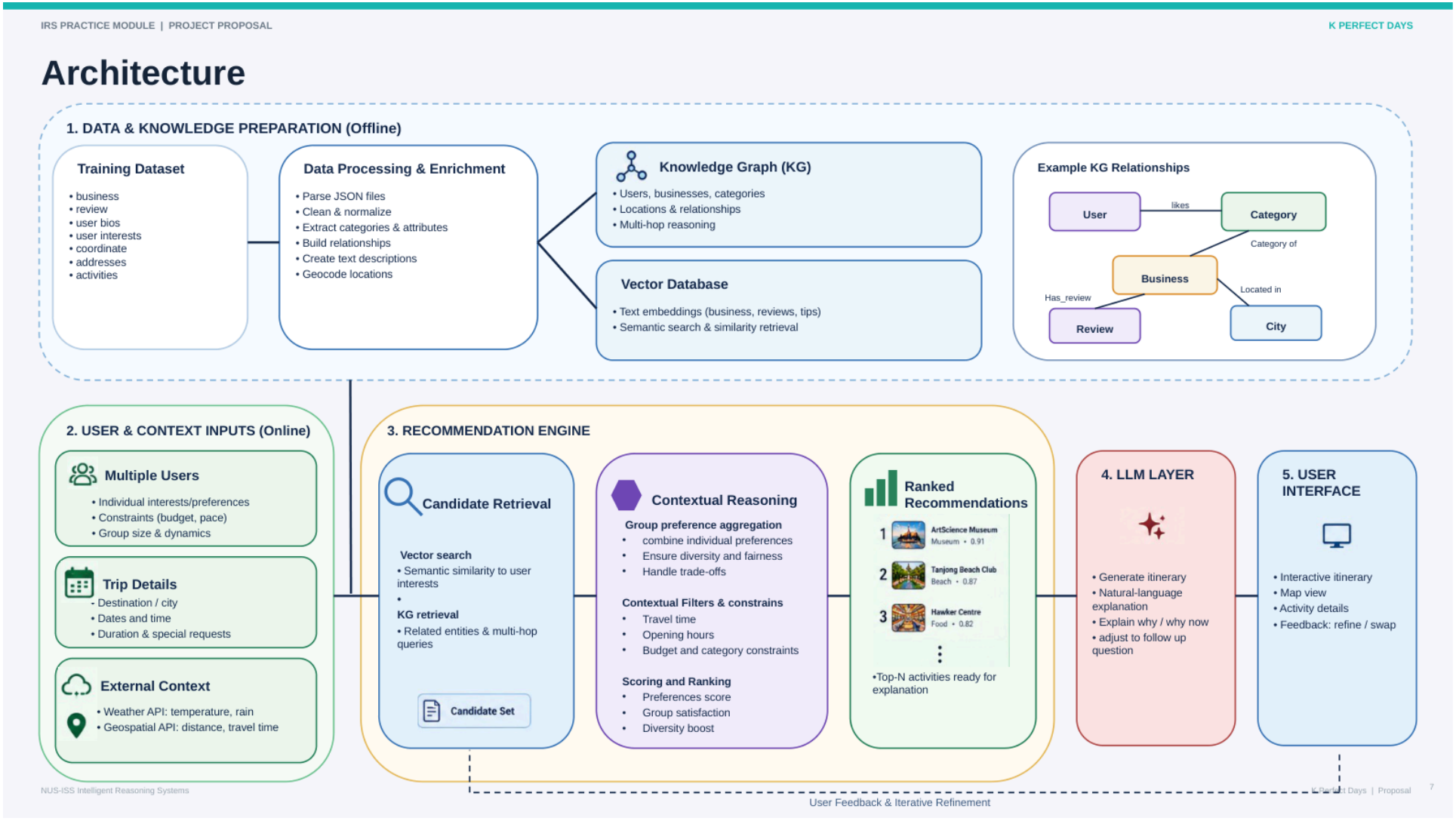
Tourists attractions and activities dataset from kaggle. The dataset includes restaurants, shops and places to visit.



Surveys

Real user data and their interests will be collected from family and friends to validate the model's recommendations




IRS PRACTICE MODULE | PROJECT PROPOSAL
K PERFECT DAYS
Architecture
1. DATA & KNOWLEDGE PREPARATION (Offline)
Training Dataset
• business
• review
• user bios
• user interests
• coordinate
• addresses
• activities
Data Processing & Enrichment
• Parse JSON files
• Clean & normalize
• Extract categories & attributes
• Build relationships
• Create text descriptions
• Geocode locations
Knowledge Graph (KG)
• Users, businesses, categories
• Locations & relationships
• Multi-hop reasoning
Vector Database
• Text embeddings (business, reviews, tips)
• Semantic search & similarity retrieval
Example KG Relationships
User
likes
Category
Category of
Business
Located in
Has_review
Review
City
2. USER & CONTEXT INPUTS (Online)
Multiple Users
• Individual interests/preferences
• Constraints (budget, pace)
• Group size & dynamics
Trip Details
- Destination / city
• Dates and time
• Duration & special requests
External Context
• Weather API: temperature, rain
• Geospatial API: distance, travel time
3. RECOMMENDATION ENGINE
Candidate Retrieval
Vector search
• Semantic similarity to user interests
KG retrieval
• Related entities & multi-hop queries
Candidate Set
Contextual Reasoning
Group preference aggregation
• combine individual preferences
• Ensure diversity and fairness
• Handle trade-offs
Contextual Filters & constrains
• Travel time
• Opening hours
• Budget and category constraints
Scoring and Ranking
• Preferences score
• Group satisfaction
• Diversity boost
Ranked Recommendations
1 ArtScience Museum Museum • 0.91
2 Tanjong Beach Club Beach • 0.87
3 Hawker Centre Food • 0.82
•Top-N activities ready for explanation
4. LLM LAYER
• Generate itinerary
• Natural-language explanation
• Explain why / why now
• adjust to follow up question
5. USER INTERFACE
• Interactive itinerary
• Map view
• Activity details
• Feedback: refine / swap
User Feedback & Iterative Refinement
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IRS PRACTICE MODULE | PROJECT PROPOSAL

K PERFECT DAYS

Technique Used

Overall, we are utilizing knowledge-based reasoning and knowledge discovery techniques to automate recommendation to communicate it to the user using natural language

Project component	Technique / algorithm to use
User preference representation	User-Based Filtering
Text → semantic representation	Embeddings / Semantic Representation
User ↔️ activity similarity	Cosine Similarity
Recommendations from similar users	Collaborative Filtering (CF)
User–item interaction patterns	Association Rule Mining / Apriori
Predict suitable activity/category	Classification Rules
Explicit requirements	Knowledge-Based Recommendation
Weather, budget, opening time, distance	Constraint-Based Reasoning
Combine several criteria	Multi-Criteria Decision Making (MCDM)
Graph relationships	Knowledge Graph / Semantic Representation
Reason across relationships	Graph Traversal / Multi-hop Reasoning
KG semantic similarity	Knowledge Graph Embeddings
Basic KG embedding	TransE / TransR
More expressive KG embeddings	DistMult / ComplEx
Final recommendation ordering	Top-K Recommendation

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Evaluation Metrics

Evaluation area	Technique / metric	What it tells you
Preference prediction	Holdout / Leave-One-Out	Can CF recover a known user preference that you deliberately hid?
	MAE	How far predicted ratings are from actual ratings
Retrieval	Precision@K	Of the Top-K predicted likes, how many were actually relevant?
	Recall@K	Of everything the user actually liked, how much did Top-K recover?
	Hit Rate@K (optional)	Did the hidden liked item appear anywhere in Top-K?
Ranking	NDCG@K	Did highly relevant preferences appear near the top?
	MRR (optional)	How high did the first correct hidden preference appear?
Group quality	Average Satisfaction	Overall satisfaction across group members
	Least-Misery / Min Satisfaction	Is one person being consistently ignored?
	Fairness	Are recommendations reasonably balanced among members?
Real user feedback	Like/dislike, rating, accept/reject	Explicit feedback on recommendation quality
	Click / save / select / visit	Implicit behavioral feedback
	A/B Testing	Compare two recommender approaches with real users

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K PERFECT DAYS

Challenges & Room of Improvements

Challenges

- limited open source datasets leaving us with domain mismatch
- inconsistent data labeling
- compute challenges with open source models
- implicit vs. explicit preferences might face some confidence inconsistency

Room of Improvements

- Adding time slots from recommendation can improve the quality of results
- Adding data set for behavioural preferences will be efficient *(such as suggest lunch after a day on the beach)*
- Train diverse dataset to eliminate biases
- For better recommendation, activities dataset is required

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